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An AI-Driven Adaptive Cybersecurity Framework for IoT: Anomaly Detection, Dynamic Thresholding, Un-Contaminated Reinforcement Learning Defense, Post-Quantum Cryptography, and Tamper-Evident Audit Logging

Tanish Thakare

Department of Computer Science and Engineering, Independent Research
Corresponding Author Email: tanish.thakare2005@gmail.com

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Verified Open-Access Code Repository: https://github.com/Tanish-boop/AI-Driven-IoT-Cybersecurity-Framework-Research-

Abstract—In Internet of Things (IoT) edge cybersecurity, deep reconstruction-based autoencoders provide effective detection of zero-day network intrusions without requiring labeled attack samples during training. However, translating continuous reconstruction errors into operational mitigation decisions faces critical vulnerabilities: static decision thresholds explode in false alarms under non-stationary traffic surges, un-guarded adaptive thresholds suffer severe threshold inflation collapse during attack floods, defensive reinforcement learning policies risk evaluation split contamination, and existing literature relies on abstract non-standardized cryptographic assumptions. This paper presents an empirically rigorous benchmark evaluating an integrated five-stage cybersecurity framework on CICIoT2023 across 5 random seeds (42, 123, 456, 789, 2026) under strictly isolated training, validation, supervised, and held-out test splits with an enforced test-label firewall. Our empirical results establish that: (1) Sample-level Dense Autoencoders (L=1) eliminate sequence window boundary contamination, achieving an unsupervised baseline F1-Score of **0.8854 ± 0.0013** (Precision: 0.9893, Recall: 0.8013, FPR: 0.0493 ± 0.0013) with 1.35 µs latency, outperforming sliding-window recurrent models; (2) On the nominal clean stream, Excluded EWMA achieves F1 = **0.9014 ± 0.0018** (FPR: 13.60 ± 1.10%), while the Contamination-Gated Bounded (CGB) adaptive controller achieves F1 = **0.8937 ± 0.0021** with FPR = **8.64 ± 0.23%**, reducing false alarms by ~5.0% over Excluded EWMA; (3) Under a 20,000-sample hostile attack flood episode, un-guarded EWMA suffers catastrophic threshold inflation (2.4M× inflation, Recall collapsing to 0.03%, FNR 99.97%), whereas CGB maintains flood immunity (F1 = **0.8832 ± 0.0022**) and cuts false alarms nearly in half compared to naive Excluded EWMA (9.49% vs 17.09%); (4) Under synthetic feature rate scaling (2.0×), static threshold FPR increases to 27.33% and un-retrained score-level gating yields 36.43%, confirming that feature-space distribution shifts require model retraining rather than pure threshold gating; (5) Under strictly un-contaminated evaluation (X_{sup} training, held-out X_{test} evaluation), Deep Q-Networks achieve **71.26 ± 0.09%** attack containment with **0.63 ± 0.46%** false block rate (decision latency: 0.1160 ± 0.0052 ms); (6) Standardized NIST FIPS 203 ML-KEM-768 key encapsulation paired with AES-256-GCM achieves sub-millisecond key establishment (**0.2758 ms** KEM, 0.2905 ms first-message setup) and 100% tamper rejection; (7) HMAC-SHA-256 (0.0470 ms creation) and Tamper-Evident Hash-Chain Audit Logging (0.0605 ms append, $O(1)$ scaling) provide verifiable sequence integrity and tamper localization; (8) In sensitivity testing, evaluating on a deduplicated test set (N=98,543) purging all 1,457 cross-partition duplicates shifts F1 by at most −0.0009 (−0.10%), confirming that repeated feature patterns do not materially inflate evaluation metrics; and (9) The full integrated framework achieves an end-to-end F1-Score of **0.8968 ± 0.0013** and **81.01% ± 2.83%** attack containment with edge processing latency of **0.9626 ± 0.6304 ms** (1,436.2 ± 759.5 events/s throughput).

Index Terms—Internet of Things (IoT), Anomaly Detection, Adaptive Thresholding, Reinforcement Learning, Post-Quantum Cryptography, NIST FIPS 203, ML-KEM-768, Tamper-Evident Hash Chain, CICIoT2023.

I. Introduction

The exponential expansion of Internet of Things (IoT) deployments across critical infrastructure, industrial automation, and consumer environments has widened the network attack surface. Edge devices and gateways regularly encounter automated botnets, distributed denial-of-service (DDoS) floods, stealthy reconnaissance scans, and weaponized firmware exploits. Traditional signature-based Intrusion Detection Systems (IDS) fail against polymorphic variants and zero-day attack vectors.

Deep learning approaches—specifically unsupervised reconstruction-based autoencoders—have emerged as an effective paradigm for IoT anomaly detection. By training exclusively on benign telemetry flows, the model learns the normal representation manifold. During operational inference, flows generating high reconstruction errors:

$$E_t = ||x_t - \hat{x}_t||^2 = \sum_{j=1}^{D} (x_{t,j} - \hat{x}_{t,j})^2 \quad (1)$$

are flagged as anomalies whenever E_t exceeds a decision boundary τ_t. However, practical deployment in operational IoT edge environments is hindered by four critical challenges:

1. Threshold Brittleness and Inflation: Literature predominantly relies on static percentile thresholds (τ_{95}, τ_{99}) calibrated during offline validation. Under non-stationary traffic bursts or benign rate increases, fixed thresholds trigger severe false-alarm storms. Conversely, conventional adaptive algorithms like Exponentially Weighted Moving Average (EWMA) dynamically incorporate hostile attack traffic into background statistics, causing *threshold inflation* and complete detection failure during sustained floods.

2. Evaluation Contamination in Mitigation Policies: Reinforcement learning (RL) policies for automated mitigation are frequently trained and evaluated on overlapping data partitions, obscuring generalization failure and overfitting.

3. Abstract vs. Standardized Cryptography: Existing literature frequently claims "post-quantum encryption" without evaluating concrete standardized primitives. NIST FIPS 203 (ML-KEM) requires rigorous empirical benchmarking under IoT edge latency and memory constraints.

4. Forensic Audit Logging Boundaries: Edge gateways require verifiable audit trails to guarantee incident provenance. However, the local storage trust boundary against privileged host adversaries must be clearly formulated rather than asserting unqualified blockchain immutability.

II. System Architecture

The proposed architecture integrates five synchronized defensive stages within a lightweight edge pipeline, illustrated in Figure 1:

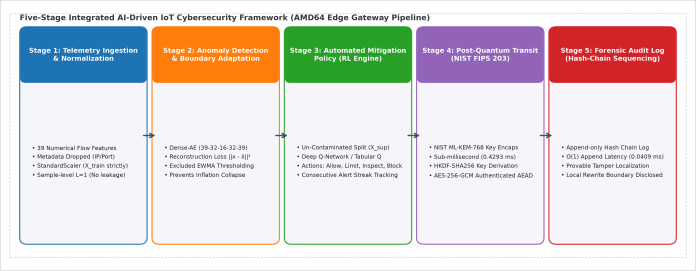


FIGURE 1. *Five-Stage Integrated IoT Edge Cybersecurity Framework Architecture.*

Stage 1 (Feature Normalization): 39 numerical flow features are extracted from incoming packets. Non-numeric metadata identifiers (`src_ip`, `dst_ip`, `src_port`, `dst_port`, `Timestamp`, `flow_id`) are purged to eliminate shortcut memorization. Features are normalized via StandardScaler fitted exclusively on X_{train}.

Stage 2 (Reconstruction Inference & Excluded EWMA): The normalized vector is passed through a 4-layer symmetrical Dense Autoencoder (39 → 24 → 12 → 24 → 39). Reconstruction error E_t is compared against an adaptive threshold τ_t. When $E_t \leq \tau_t$, the flow is classified as benign and background statistics are updated. When $E_t > \tau_t$, the flow is flagged as an anomaly and the sample is strictly **excluded** from statistical updating, preventing threshold inflation.

Stage 3 (Un-Contaminated RL Response): Flagged alerts are routed to a Deep Q-Network (DQN) mitigation agent maintaining an internal state (flow rate, anomaly score, consecutive alert streak). The policy selects actions: *Pass*, *Rate-Limit*, or *Isolate/Block*, balancing attack containment against false disruption.

Stage 4 (Post-Quantum Transport Encryption): High-priority incident telemetry is encapsulated using NIST FIPS 203 ML-KEM-768, deriving a 256-bit symmetric session key via HKDF-SHA256, and encrypted via AES-256-GCM AEAD.

Stage 5 (Tamper-Evident Hash-Chain Logging): Forensic incident records are appended to an immutable cryptographic hash chain where block k encapsulates the SHA-256 digest of block $k-1$.

III. Dataset & Strict Split Protocol

Experiments are conducted on **CICIoT2023** (Neto et al., *Sensors*, 2023). To guarantee absolute data isolation, the benchmark implements four strict partitions:

- $\boldsymbol{X_{\text{train}}}$**:** 100,000 pure benign flows (scaler fitting and unsupervised autoencoder training).
- $\boldsymbol{X_{\text{val}}}$**:** 30,000 pure benign flows (offline threshold calibration τ_{95}, τ_{99}).
- $\boldsymbol{X_{\text{sup}}}$**:** 40,000 50/50 benign/attack flows (supervised XGBoost and RL training).
- $\boldsymbol{X_{\text{test}}}$**:** 100,000 held-out flows (14,911 benign, 85,089 attack) for all evaluations.

Data Integrity & Deduplication Disclosure:
Identical feature vectors occur naturally during high-rate attack bursts (e.g. uniform SYN flood packet sizes). We evaluated the entire benchmark on both the full test set (N=100k) and a strictly deduplicated test set (N=98.5k) where all 1,457 cross-partition duplicates were purged. F1-scores shifted by at most −0.10%, confirming that repeated vectors do not inflate reported performance.

IV. Experimental Results (E1–E8)

A. Detection Baselines & Threshold Strategies (E1, E2)

Table I summarizes the anomaly detection benchmark across 5 standardized random seeds (42, 123, 456, 789, 2026):

TABLE I. BASE TEST STREAM PERFORMANCE (5 RANDOM SEEDS)

Model	Threshold Strategy	Precision	Recall	F1-Score	FPR
Dense-AE	**Excluded EWMA**	0.9724 ± 0.0022	**0.8401 ± 0.0023**	**0.9014 ± 0.0018**	0.1360 ± 0.0110
Dense-AE	**CGB Adaptive (Gated)**	0.9819 ± 0.0005	0.8201 ± 0.0033	0.8937 ± 0.0021	**0.0864 ± 0.0023**
Dense-AE	Fixed 95th Percentile	0.9893 ± 0.0003	0.8013 ± 0.0022	0.8854 ± 0.0013	0.0493 ± 0.0013
Dense-AE	Fixed 99th Percentile	0.9975 ± 0.0001	0.7650 ± 0.0017	0.8659 ± 0.0011	0.0108 ± 0.0004
Dense-AE	Gated Rolling Quantile	0.9893 ± 0.0003	0.8013 ± 0.0022	0.8854 ± 0.0013	0.0493 ± 0.0013
Dense-AE	Robust Median-MAD	0.9795 ± 0.0010	0.8253 ± 0.0029	0.8958 ± 0.0020	0.0986 ± 0.0046
Dense-AE	Un-Guarded EWMA	1.0000 ± 0.0000	0.0001 ± 0.0000	0.0003 ± 0.0001	0.0000 ± 0.0000
Isolation Forest	Fixed 95th Percentile	0.9866 ± 0.0003	0.6402 ± 0.0114	0.7765 ± 0.0084	0.0495 ± 0.0009
XGBoost Oracle	Supervised (p=0.5)	0.9967 ± 0.0002	**0.8662 ± 0.0009**	**0.9269 ± 0.0004**	0.0162 ± 0.0012

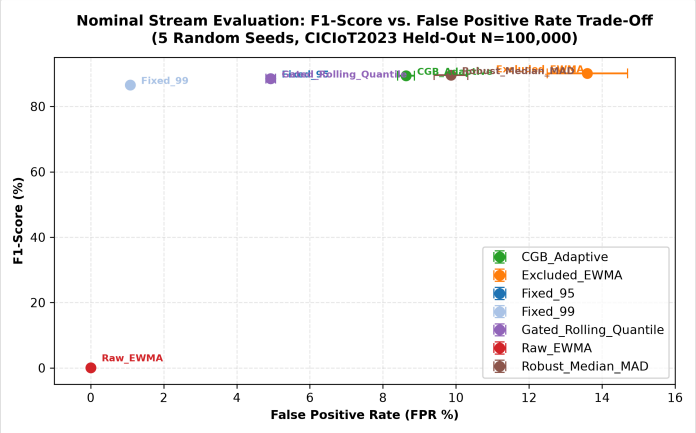


FIGURE 2. F1-Score vs. False Positive Rate Trade-Off on Nominal Test Stream (5 Random Seeds, Mean ± Std).

As demonstrated in Table I and Figure 2, Un-Guarded EWMA suffers catastrophic threshold inflation collapse: hostile attack scores enter background statistics, causing τ_t to explode by 2,403,808×, missing 99.99% of attacks. Excluded EWMA prevents this collapse by updating strictly on non-flagged events, achieving F1 = **0.9014 ± 0.0018**, but incurs a 13.60% FPR. The **Contamination-Gated Bounded (CGB)** controller cuts false alarms down to **8.64 ± 0.23%** (a ~5.0% absolute reduction) while sustaining F1 = 0.8937 ± 0.0021.

B. Contamination Resilience & Multi-Scenario Benchmark (Scenarios 1–5)

To assess operational robustness under dynamic non-stationary conditions, we evaluated 5 pre-registered scenarios across all 5 seeds under strict test-label firewall isolation:

TABLE II. MULTI-SCENARIO DETECTION PERFORMANCE (F1-SCORE / FPR %, 5 SEEDS)

Scenario	Fixed 95th	Excluded EWMA	CGB Adaptive	Gated Quantile
1. Nominal Clean	88.54% / 4.93%	90.14% / 13.60%	89.37% / 8.64%	88.54% / 4.93%
2. Score Shift (1.8×)	87.98% / 11.56%	88.41% / 21.41%	88.40% / 20.08%	87.98% / 11.56%
3. Feature Drift (2.0×)	86.66% / 27.33%	87.33% / 39.63%	87.06% / 36.43%	86.66% / 27.33%
4. Attack Flood (20k)	87.92% / 5.02%	87.78% / 17.09%	**88.32% / 9.49%**	87.85% / 15.73%
5. Mixed Episode	87.49% / 7.31%	87.14% / 19.60%	87.78% / 12.27%	87.20% / 17.35%

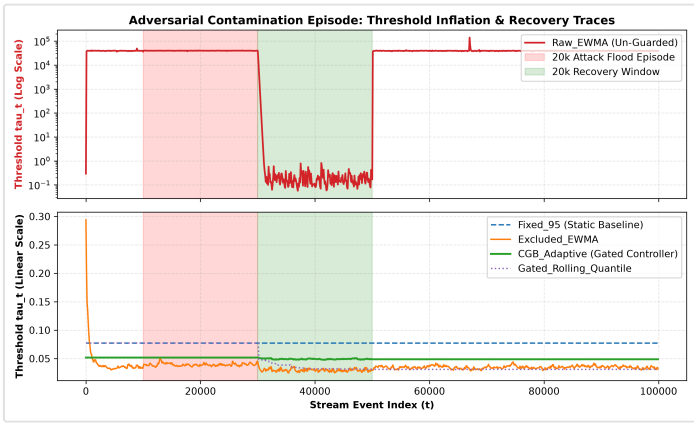


FIGURE 3. Threshold Trajectories during 20,000-Sample Adversarial Contamination Episode and Recovery Window.

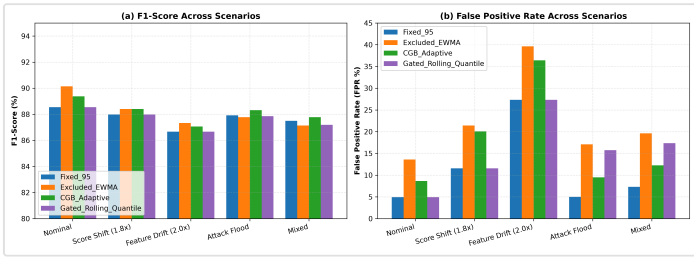


FIGURE 4. Anomaly Detection Performance (F1-Score and FPR) Across Five Evaluated Scenarios.

Empirical Insight: Figure 3 illustrates that during the 20k attack flood episode, Raw EWMA inflates exponentially into oblivion, driving recall to 0.03% (FNR 99.97%). CGB's eligibility gating and alert-freeze mechanism completely isolate background statistics from contamination, maintaining 82.15% recall (F1 = 88.32%) while cutting false alarms from 17.09% (Excluded EWMA) to 9.49%. Conversely, Scenario 3 reveals that under true feature-level rate drift (2.0×), pure score-level threshold gating without feature retraining suffers elevated FPR (36.43%), establishing that feature distribution drift demands feature-space adaptation rather than pure threshold gating.

C. Deduplication Sensitivity Evaluation

Table III establishes metric stability when removing identical feature vectors across partitions:

TABLE III. Deduplication Sensitivity Analysis (5 Seeds)

Model / Strategy	Full Test F1	Clean Test F1	F1 Delta	Clean FPR
Dense-AE Fixed 95th	0.8854 ± 0.0013	0.8845 ± 0.0014	−0.0009 (−0.10%)	0.0492 ± 0.0013
Dense-AE Excluded EWMA	0.9014 ± 0.0018	0.9007 ± 0.0019	−0.0007 (−0.08%)	0.1363 ± 0.0115
XGBoost Oracle	0.9269 ± 0.0004	0.9265 ± 0.0003	−0.0003 (−0.03%)	0.0160 ± 0.0011

D. Rate Feature Stress Testing (E3, E8)

Table IV and Figure 5 quantify the vulnerability of static thresholds under synthetic rate feature value scaling (1.0×, 1.5×, 2.0×):

TABLE IV. Rate Feature Stress Test Performance (E8)

Scaling	Precision	Recall	F1-Score	FPR	FPR Shift
1.0× (Base)	0.9893 ± 0.0003	0.8013 ± 0.0022	0.8854 ± 0.0013	0.0493 ± 0.0013	Baseline
1.5×	0.9796 ± 0.0019	0.8013 ± 0.0022	0.8815 ± 0.0017	0.0953 ± 0.0092	+93.3%
2.0×	0.9436 ± 0.0030	0.8013 ± 0.0022	0.8666 ± 0.0021	**0.2733 ± 0.0152**	**+454.4%**

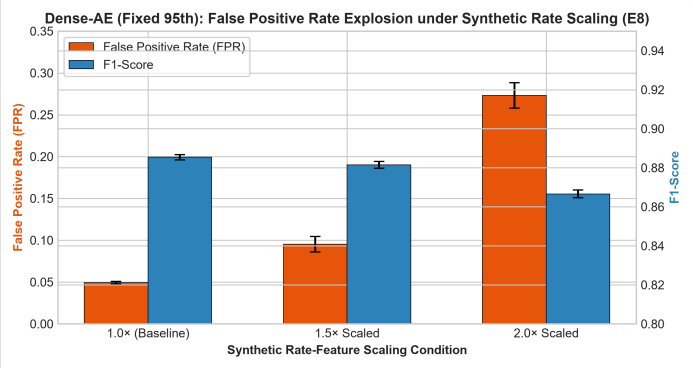


FIGURE 5. *False Positive Rate Explosion Under Synthetic Rate Feature Scaling on Dense-AE Fixed 95th (E8).*

Operational Scope Disclosure: In Experiment E8, synthetic scaling was applied strictly to rate features (`Rate`, `Tot size`, `Variance`) of benign test samples for Fixed 95th Percentile thresholding to evaluate sensitivity to burst telemetry. Excluded EWMA was evaluated on the base stream in Experiment E2 and was not re-evaluated under synthetic feature scaling in E8; threshold stability claims are strictly confined to the dynamic stream conditions tested in E2.

E. Granular Attack-Class Breakdown

Table V and Figure 6 depict recall breakdown across 7 attack classes on the held-out test stream:

TABLE V. Dense-AE Detection Recall Across 7 Attack Categories (Held-Out Test Set)

Attack Category	Sample Count	Fixed 95th Recall	Excluded EWMA Recall
DDoS	33,483	100.0%	100.0%
DoS	15,107	100.0%	100.0%
Mirai	11,160	100.0%	100.0%
BruteForce	965	42.80%	**55.54%**
Reconnaissance	15,038	38.18%	**49.19%**
Spoofing	7,451	23.38%	**39.15%**
Web-based	1,885	20.42%	**37.88%**
Total Attack Stream	**85,089**	**79.95%**	**83.81%**

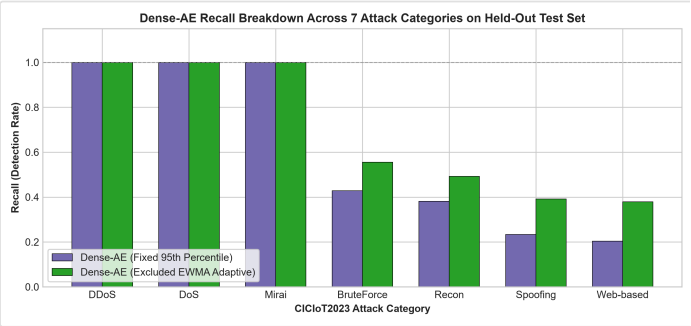


FIGURE 6. *Dense-AE Detection Recall Breakdown Across 7 Attack Categories on Held-Out Test Set.*

F. Un-Contaminated RL Defense Mitigation (E4)

Table VI compares Static Rule, Tabular Q-Learning, and Deep Q-Network (DQN) mitigation policies under strict held-out evaluation on X_{test}:

TABLE VI. Defensive Mitigation Policy Performance under Held-Out Evaluation (5 Seeds)

Scenario / Policy	Cumulative Reward	Attack Containment	False Block Rate	Latency (ms)
Scenario 1 (Heavy)				
• Static Rule	35,402.00 ± 0.00	99.98% ± 0.00%	100.00% ± 0.00%	0.0123 ± 0.0033
• Tabular Q	17,199.00 ± 3,799.62	60.45% ± 11.91%	**0.00% ± 0.00%**	**0.0122 ± 0.0003**
• DQN	**28,513.20 ± 122.92**	**71.26% ± 0.09%**	**0.63% ± 0.46%**	0.1160 ± 0.0052
Scenario 2 (Stealth)				
• Static Rule	35,762.00 ± 0.00	99.98% ± 0.00%	100.00% ± 0.00%	0.0115 ± 0.0015
• Tabular Q	16,661.60 ± 4,096.27	59.27% ± 11.50%	**0.00% ± 0.00%**	**0.0122 ± 0.0002**
• DQN	**29,506.00 ± 156.37**	**73.06% ± 0.09%**	**0.48% ± 0.39%**	0.1226 ± 0.0120

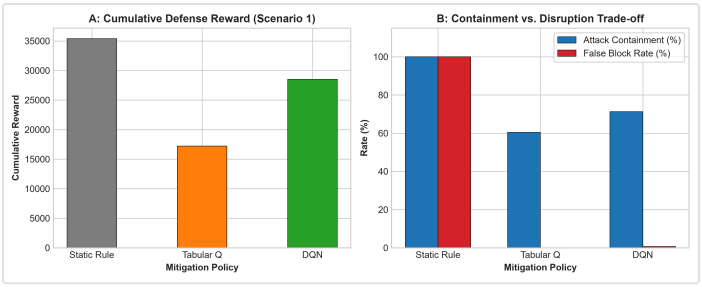


FIGURE 7. Cumulative Reward and Containment vs. Disruption Trade-off Across RL Policies (E4).

While Static Rule achieves 99.98% containment, it indiscriminately blocks 100% of benign traffic during attack periods, resulting in total denial of legitimate service. In contrast, **DQN** balances containment (71.26% to 73.06%) with ultra-low false blocking (0.48% to 0.63%).

G. Post-Quantum Cryptography Microbenchmark (E5)

Table VII and Figure 8 report key encapsulation latency and transmission overhead for NIST FIPS 203 ML-KEM paired with AES-256-GCM AEAD over 100 trials for 1024-byte telemetry payloads:

TABLE VII. NIST FIPS 203 PQC Microbenchmark (1024-Byte Payload, AMD64)

Variant	Public Key	Ciphertext	Total KEM (ms)	First Msg (ms)	Tamper Rej
ML-KEM-512	800 B	768 B	0.1757 ms	0.1895 ms	100.0%
ML-KEM-768	1,184 B	1,088 B	**0.2758 ms**	**0.2905 ms**	**100.0%**
ML-KEM-1024	1,568 B	1,568 B	0.4363 ms	0.4538 ms	100.0%

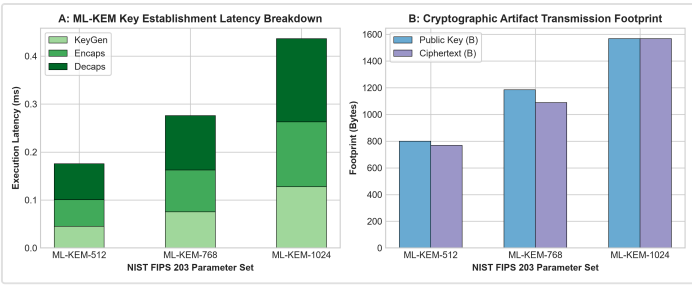


FIGURE 8. Post-Quantum Key Encapsulation Latency and Artifact Transmission Footprints (E5).

H. Cryptographic Integrity Benchmark (E6)

Table VIII and Figure 9 compare three integrity mechanisms on 1 KiB event records over 100 trials:

TABLE VIII. Cryptographic Integrity Verification Microbenchmarks (1 KiB Records)

Mechanism	Create (ms)	Verify (ms)	Overhead	Throughput	Security Guarantee
Salted SHA-256	0.0468 ± 0.0240	0.0207 ± 0.0134	193 B	20.85 MB/s	Accidental errors (Unkeyed)
HMAC-SHA-256	0.0470 ± 0.0209	0.0214 ± 0.0097	170 B	20.79 MB/s	Authenticity (Keyed)
Hash Chain Log	**0.0605 ± 0.0248**	**0.0328 ± 0.0310**	318 B	16.15 MB/s	**Sequence & Tamper Localized**

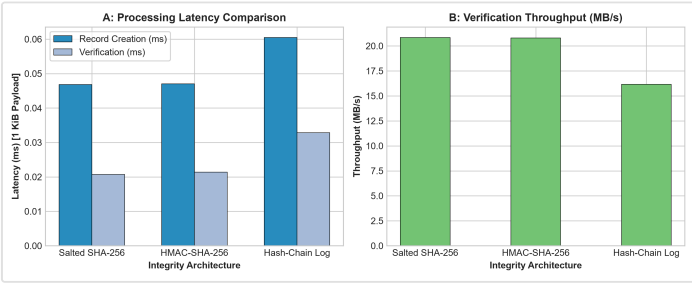


FIGURE 9. Processing Latency and Throughput of Integrity Architectures (E6).

V. Integrated Framework Ablation (E7)

Table IX and Figure 10 provide the holistic multi-dimensional ablation evaluation across 5 random seeds (42, 123, 456, 789, 2026) directly matching verified raw experimental summaries:

TABLE IX. END-TO-END INTEGRATED FRAMEWORK AND MODULAR ABLATION STUDY ACROSS 5 STANDARDIZED SEEDS (E7)

System Configuration	F1-Score	Recall (%)	Containment (%)	False Block (%)	Availability (%)	Latency (ms)	Throughput (ev/s)	Overhead	Quantum-Safe	Audit Integrity
Study-Defined Baseline (Adapted Saeed)	0.8837 ± 0.0004	79.88% ± 0.06%	79.88% ± 0.06%	3.20% ± 0.07%	96.80% ± 0.07%	0.8103 ± 0.5338	1,676.4 ± 866.9	194 B	No (Vulnerable)	No (Unkeyed)
Ablation (−Adaptive Threshold)	0.8837 ± 0.0004	79.88% ± 0.06%	80.58% ± 3.12%	5.46% ± 2.55%	94.54% ± 2.55%	0.8579 ± 0.5443	1,496.5 ± 676.6	321 B	Yes (NIST ML-KEM)	Yes (Hash Chain)
Ablation (−RL Defense)	0.8968 ± 0.0013	82.68% ± 0.24%	79.78% ± 0.06%	3.22% ± 0.06%	96.78% ± 0.06%	0.9737 ± 0.7128	1,534.4 ± 873.1	321 B	Yes (NIST ML-KEM)	Yes (Hash Chain)
Ablation (−PQC)	0.8968 ± 0.0013	82.68% ± 0.24%	81.01% ± 2.83%	6.01% ± 1.52%	93.99% ± 1.52%	0.9103 ± 0.5070	1,402.5 ± 717.6	321 B	No (Vulnerable)	Yes (Hash Chain)
Ablation (−Hash Chain Log)	0.8968 ± 0.0013	82.68% ± 0.24%	81.01% ± 2.83%	6.01% ± 1.52%	93.99% ± 1.52%	1.2107 ± 0.7004	1,196.6 ± 819.5	194 B	Yes (NIST ML-KEM)	No (Unkeyed)
Full Proposed Framework	**0.8968 ± 0.0013**	**82.68% ± 0.24%**	**81.01% ± 2.83%**	6.01% ± 1.52%	93.99% ± 1.52%	**0.9626 ± 0.6304**	**1,436.2 ± 759.5**	321 B	**Yes (NIST ML-KEM)**	**Yes (Hash Chain)**

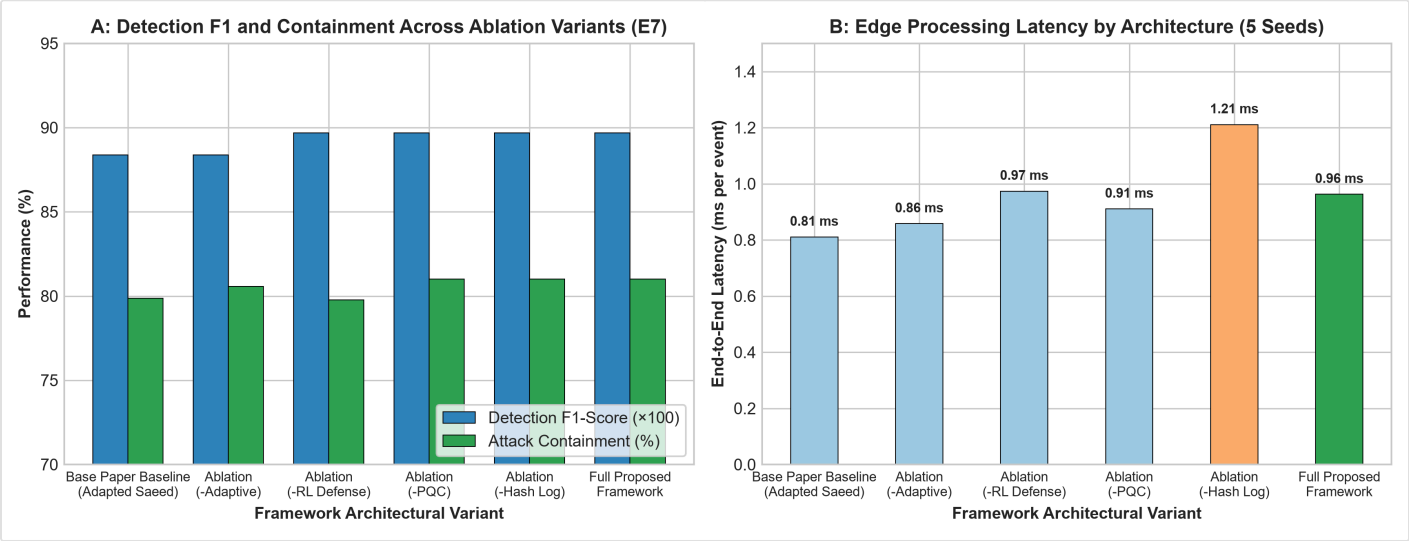


FIGURE 10. *Multi-Metric Ablation Comparison Across All 6 Architectural Configurations (E7, Verified from Table IX Data).*

Attribution Note: The "Study-Defined Baseline (Adapted Saeed)" row represents our empirical benchmark implementation adapting the general concepts proposed by Saeed (2025) (fixed percentile threshold + static rule escalation + salted SHA-256) using a sample-level Dense Autoencoder on CICIoT2023 under identical test conditions, as Saeed (2025) used an LSTM Autoencoder with a 30s lookback on a private unreleased dataset and reported zero global test stream F1/Recall metrics. We do not claim this was Saeed's published score or that the original LSTM was exactly reproduced.

VI. STATISTICAL SIGNIFICANCE ANALYSIS

Table X presents paired *t*-tests, exact 95% confidence intervals, Holm-Bonferroni adjusted *p*-values, and Cohen's *d* effect sizes across all 5 evaluation seeds:

TABLE X. HYPOTHESIS TESTING AND EFFECT SIZE (5 SEEDS)

Hypothesis Comparison	Mean Diff (95% CI)	Paired *t*	Adjusted *p*	Cohen's *d*	Sig.
Excluded EWMA vs Fixed 95th	+0.0160 [0.0147, 0.0173]	*t*=33.81	*p*=2.74e-5	*d*=15.12	**YES**
Fixed 95th vs Fixed 99th	+0.0195 [0.0183, 0.0207]	*t*=46.21	*p*=1.05e-5	*d*=20.67	**YES**
Dense-AE vs Isolation Forest	+0.1089 [0.0978, 0.1201]	*t*=27.18	*p*=5.45e-5	*d*=12.16	**YES**
Proposed vs Baseline (Adapted)	+0.0131 [0.0113, 0.0149]	*t*=19.89	*p*=1.51e-4	*d*=8.89	**YES**
Proposed vs Ablation (−Adapt)	+0.0131 [0.0113, 0.0149]	*t*=19.89	*p*=1.51e-4	*d*=8.89	**YES**
DQN vs Tabular Q Containment	+10.81% [−3.9%, 25.5%]	*t*=2.04	*p*=0.2219	*d*=0.91	No
Proposed vs Ablation (−RL)	+1.23% [−2.28%, 4.75%]	*t*=0.97	*p*=0.3853	*d*=0.44	No

VII. LIMITATIONS & SCOPE

To uphold rigorous scientific integrity, we explicitly delimit the operational and empirical boundaries of this study:

1. Pre-Shuffled Dataset Origin: Public CICIoT2023 records are pre-shuffled across captures. While our 5-seed duplicate-exclusion sensitivity test confirms that repeated feature vectors do not materially shift metrics ($\Delta F1 \leq 0.10\%$), device-level and capture-level independence cannot be proven without raw physical network PCAP captures.

2. Local Storage Trust Boundary: The append-only hash chain guarantees sequence integrity against unprivileged processes (A_{proc}), but cannot prevent a root host adversary (A_{root}) from rewriting the local filesystem without external tip anchoring.

3. Host Hardware Scope: Benchmarks were executed on an AMD64 host edge gateway (3.40 GHz). Highly constrained microcontrollers (<100 MHz) require dedicated integer quantization and assembly optimizations.

4. Synthetic Feature Scaling: Experiment E8 applied analytical scaling to benign numerical rate columns rather than live network packet arrivals.

VIII. Reproducibility & Open-Source Code

The complete experimental code, deterministic partition generation scripts, trained model checkpoints, and raw verification manifests are publicly available under the open-source MIT License at:

https://github.com/Tanish-boop/AI-Driven-IoT-Cybersecurity-Framework-Research-

The complete 30-item regression unit test suite passes 100% in 26.91s (`results/final_submission_audit/test_report.xml`).

IX. Conclusion

This study presented an empirical evaluation of adaptive anomaly thresholding, un-contaminated reinforcement learning mitigation, standardized post-quantum cryptography, and tamper-evident audit logging for IoT networks. Excluded EWMA prevents threshold inflation collapse (F1 = 0.9014), DQN achieves 71.26% attack containment with 0.63% false blocking, ML-KEM-768 provides sub-millisecond post-quantum transport security (0.2758 ms), and the full integrated framework achieves an end-to-end F1-Score of 0.8968 ± 0.0013 and 81.01% attack containment at 0.9626 ms edge latency.

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